Chapter 13

Personal Health Care

1. Healthy Skin, Hair, and Nails
2. Healthy Teeth and Mouth
3. Healthy Eyes and Ears

Lesson 1 — Healthy Skin, Hair, and Nails

Your Skin — Your Largest Organ

Your skin consists of two main layers: the **epidermis** (outer, thinner layer of living and dead cells) and the **dermis** (thicker inner layer containing blood vessels, nerves, and glands). Beneath both sits the subcutaneous layer — a cushion of fat and connective tissue.

Your skin has three critical functions: Protection (barrier against pathogens and damage), Temperature regulation (blood vessels dilate or constrict; sweat glands cool the body), and Sensation (nerve receptors detect pressure, pain, heat, and cold). The dermis also contains **sebaceous glands** that produce sebum — an oily substance that keeps skin and hair moisturized.

UV Protection — Updated Guidelines

A suntan is NOT a sign of health — it's evidence of skin damage. Use sunscreen with **SPF 30 or higher** (updated from older SPF 15 recommendations) on all exposed skin, even on cloudy days. Apply 15–30 minutes before going outside and reapply every 2 hours.

- Use broad-spectrum (UVA + UVB) protection
- Seek shade between 10 a.m. and 4 p.m.
- Never use tanning beds — using one before age 35 increases melanoma risk by 75%
- Wear UV-blocking sunglasses

Melanoma — The ABCDE Rule

A — Asymmetry	One half doesn't match the other
B — Border	Irregular, ragged, or blurred edges

C — Color	Uneven color, or intensely black with bluish tint
D — Diameter	Wider than a pea (~6mm)
E — Evolving	Any change in size, shape, color, or new symptom (bleeding, itching) — NEW since original textbook

> Report any suspicious moles to a dermatologist immediately. Early detection of melanoma saves lives.

Common Skin Problems

- **Acne** — Clogged pores infected by bacteria. Treat by washing gently twice daily, using OTC treatments with benzoyl peroxide or salicylic acid; avoid picking.
- **Dermatitis/Eczema** — Inflamed or scaly patches from allergic reactions. Keep moisturized; may need prescription treatment.
- **Melanoma** — Most serious, potentially fatal form of skin cancer. Monitor moles using ABCDE rule. Early detection = excellent survival rate.

Lesson 2 — Healthy Teeth and Mouth

Your teeth break food into pieces small enough to digest, support your face shape, and help you speak clearly. Each tooth has a crown (covered with enamel — the hardest substance in the body), neck (at the gumline), and root (anchored in the jaw).

Protecting Your Teeth

- **Brush after every meal** — Removes plaque before acids form cavities
- **Floss daily** — Cleans between teeth where a toothbrush can't reach
- **Dentist every 6 months** — Professional cleaning and checkup
- **Wear a mouth guard** — During contact sports — dental injuries are common and expensive
- **Avoid all tobacco** — Stains teeth, causes gum recession, dramatically increases oral cancer risk

Lesson 3 — Healthy Eyes and Ears

Your Eyes

Light enters through the **cornea** (transparent front), passes through the **pupil**, is focused by the **lens**, and hits the **retina** (containing rods for dim light and cones for color vision). The optic nerve carries signals to the brain.

Screen Time — The 20-20-20 Rule: Every 20 minutes, look at something 20 feet away for 20 seconds to reduce eye strain.

Your Ears

Earbuds and headphones are a major risk factor for hearing damage. Keep volume at 60% or below; limit listening to 60 minutes at a time. Damage above 85 decibels is cumulative and **permanent**.

> Tinnitus — ringing or buzzing in the ears with no external source — is an early warning sign of hearing damage. Never ignore it after loud noise exposure.

Chapter Vocabulary

Lesson 1

Epidermis	The outer, thinner layer of skin composed of living and dead cells
Dermis	The thicker inner layer of skin containing blood vessels, nerves, and glands
Melanin	The pigment that gives skin, hair, and eyes their color and provides some UV protection
Sebaceous glands	Structures in the dermis that produce sebum (an oily secretion that moisturizes skin and hair)
Hair follicles	Sacs or cavities that surround the roots of hairs
Melanoma	The most serious and potentially fatal form of skin cancer

Lesson 2

Periodontium	The structures immediately surrounding a tooth, including the gum, periodontal ligaments, and jawbone
Plaque	A sticky film of bacteria and food particles that adheres to teeth and produces acids that cause decay
Halitosis	Bad breath, caused by food, poor oral hygiene, bacteria, or gum disease
Periodontal disease	Inflammation and infection of the gum and supporting structures of the teeth

Lesson 3

Sclera	The tough white outer layer of the eye
Cornea	The transparent front portion of the eye that bends and focuses incoming light
Retina	The inner layer of the eye containing light-sensitive rods and cones
Tinnitus	A ringing, buzzing, or hissing sound heard in the ear with no external source; often a sign of noise-induced hearing damage